First Examination 2026
Artificial Intelligence and Machine Learning

You are required to submit the following files to the designated Google Drive. 1. A pdf file containing your solutions and report for every problem. The file name should be "report.pdf". In the report file you should follow the format that has been introduced in the class. 2. A pdf file containing your source code for every problem. The file name should be "source.pdf". 3. The necessary testing files requested in every problem. **You should upload all files to your account in Google Drive no later than 14:00 pm, Oct. 10, 2026. Any files submitted later than the due time will not be accepted.** Please be reminded that this is a formal exam. You are required not to discuss with anyone and not to release any part of your solutions to others. Anyone who fails to follow the rule will be required to leave this school. Please respect yourself and your classmates.

1. Given a dataset in the file named cls1_trn.csv for training, please train a maximum likelihood estimator to learn a classifier for the data with 9 classes. Test your classifier on the testing dataset in the file named cls1_tst.csv. (1) Please calculate training accuracy and the discriminant functions $g_i(\boldsymbol{x})$, $i = 1\ldots9$. (2) Please calculate the determinant of covariance matrix in $g_i(\boldsymbol{x})$, $i = 1\ldots9$. In other words, please calculate $|\Sigma_i| \equiv \det(\Sigma_i)$, $i = 1\ldots9$. (3) Please generate a file named result1.csv to contain the testing results based on the testing data in the file cls1_tst.csv. Note that each row of the data in result1.csv should contain [testing data $\boldsymbol{x}$], [class]. (4) For the 3rd data $\boldsymbol{x}_3$ and the 10th data $\boldsymbol{x}_{10}$ in the third row and 10th row of the testing dataset in the file cls1_tst.csv, please calculate the values of $g_i(\boldsymbol{x}_3)$ and $g_i(\boldsymbol{x}_{10})$, $i = 1\ldots9$ and verify the class of $\boldsymbol{x}_3$ and $\boldsymbol{x}_{10}$, respectively. (25%)

2. Given a dataset in the file named cls2_trn.csv for training, please train a maximum likelihood estimator with the same covariance $\boldsymbol{S}$ to learn a classifier for the data with 7 classes. Test your classifier on the testing dataset in the file named cls2_tst.csv. (1) Please calculate training accuracy and the discriminant functions $g_i(\boldsymbol{x})$, $i = 1\ldots7$. (2) Please calculate the the covariance matrix $\boldsymbol{S}$. (3) Please generate a file named result2.csv to contain the testing results based on the testing data in the file cls2_tst.csv. Each row of the data in result2.csv should contain [testing data $\boldsymbol{x}$], [class]. (25%)

3. Given a dataset in the file named cls1_trn.csv as shown in problem 1 for training, please train a maximum likelihood estimator with a common diagonal covariance matrix $\boldsymbol{S}$ but unequal variances to learn a classifier for the data with 9 classes. Test your classifier on the testing dataset in the file named cls1_tst.csv. (1) Please calculate training accuracy and the discriminant

functions $g_i(\boldsymbol{x})$, $i = 1\ldots9$. (2) Please calculate the diagonal covariance matrix $\boldsymbol{S}$. (3) Please generate a file named result3.csv to contain the testing results based on the testing data in the file cls1_tst.csv. Each row of the data in result3.csv should contain [testing data $\boldsymbol{x}$], [class]. (4) Please compare your classification results with the results generated in problem 1. How many data points are with different classification results between the result1.csv and result3.csv? Please explain why they have different classification results? (25%)

4. Given a univariate dataset in the file regrs1_trn.csv for training, please train a maximum likelihood estimator to learn a regression model $g(x)$ for the data. You may use a 6-th order linear polynomial as your regression model. Test your regression model on the testing dataset in the file named regrs1_tst.csv. Please (1) show the regression model g(x). (2) calculate the coefficient of determination of your model. (3) generate a file named result4.csv to contain the testing results based on the testing data in the file regrs1_tst.csv. Each row of the data in result4.csv should contain [testing data x], [$g(x)$]. (4) If you intentionally use a 4-th order linear polynomial to learn the data that were actually generated from a 6th order polynomial, please show your 4th order polynomial. Generate a file named result4_1.csv to contain the testing results using your 4th order polynomial based on the testing data in the file regrs1_tst.csv. Compare your regression results by calculating the absolute errors of both regression results, i.e., $|g_1(\boldsymbol{x}) - g_2(\boldsymbol{x})|$ and put them into the file named result4_2.csv. (25%)

Please be reminded again. Taking problem 1 as an example, you need to submit the files as follows:

1. report1.pdf: containing your approaches solving problem 1, solution and information requested in problem 1, and your observation.
2. source1.pdf: containing your source code for problem 1.
3. result1.csv: containing testing results based on the testing data in the file cls1_tst.csv.

The exam is due at 14:00, Oct. 10, 2026.